GGG-D-641a
FEBRUARY 23, 1956
SUPERSEDING
Fed. Spec. GGG-D-641
February 22, 1943

FEDERAL SPECIFICATION

DRIFTS; DRILL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers drill drifts used for driving taper-shank tools out of machine spindles, sockets, and sleeves.

1.1.1 *Federal specification coverage.* — Federal specifications do not include all styles and sizes of the commodities indicated by the titles, but include only those generally purchased by the Federal Government.

1.2 Classification.

1.2.1 ***Type and styles.***—Drill drifts covered by this specification shall be of one type furnished in the following styles as specified (see 6.1):

Style A.—Plain.
Style B.—Flanged shoulder.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications. — There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Standards.—The following standard, of the issue in effect on date of invitation for bids, forms a part of this specification:

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. — The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of drill drifts which are otherwise in accordance with the requirements of this specification.

3.2 Material. — Drill drifts shall be drop forged of good quality tool steel.

3.3 Design.—The drill drifts shall taper 1¾ inches in 12 inches for approximately two-thirds of the entire length. One edge of the drift shall be flat along the length of the taper and the opposite edge rounded the full length of the drift. The small end of the drift shall be rounded in such a manner as to be suitable for readily driv-

362246—56—No. 202

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

ing out tools with taper shanks from taper sockets without deformation to the drift or socket. The handle end of the drift shall contain a through hole.

3.4 Hardness. — Drill drifts when tested as specified in 4.6.1, shall show a hardness as follows: The taper section from the beginning of the taper to the point end shall show a Rockwell hardness of not less than 30 nor more than 42 on the C scale. The straight or handle end shall show a Rockwell hardness of not less than 20 nor more than 30 on the C scale.

3.5 Identification marking.—All drill drifts shall be marked in a plain and permanent manner with the size, number and the manufacturer's name or trade-mark of such known character that the source of manufacture may be readily determined.

3.6 Finish. — The drill drifts shall have a smooth even surface free of rust, scale, pits, nodules, cracks, and burrs. Surfaces shall have a natural or sandblasted finish and shall be oil treated.

3.7 Style A, plain.—Style A drill drifts shall conform to table I and be similar to figure 1.

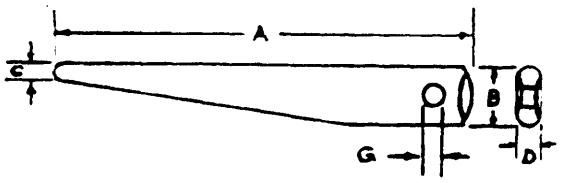


FIGURE 1.—Style A, plain drill drift.

3.8 Style B, flanged shoulder. — Style B drifts shall be provided with a flanged shoulder at the handle end. The drifts shall conform to table I and be similar to figure 2.

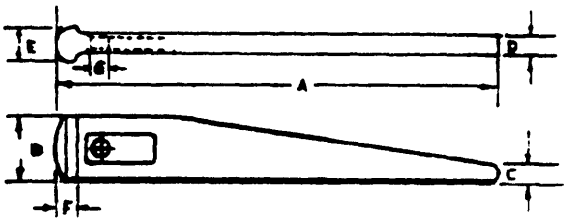


FIGURE 2.—Style B, flanged shoulder drill drift.

3.9 Workmanship.—The workmanship shall be first class in every respect and shall be free from any defect which may affect the serviceability or durability of the drill drift.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection lot. — All drill drifts of the same style and size presented at one time shall be considered a lot for purposes of acceptance, inspection, and tests.

4.2 Sampling for visual and dimensional inspection. — A random sample of drill drifts shall be taken from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements in table II in accordance with Military Standard MIL–STD–105.

TABLE I.—*Styles A and B plain and flanged shoulder drill drifts*

Size No.	Fits Morse Socket No.	Length (A)		Width of straight portion (B)		Width at small end (C)	Thickness (D)		Diameter of hole	Flange dimensions (Style B only)	
		(Min.)	(Max.)	(Min.)	(Max.)	(Max.)	(Min.)	(Max.)	(Min.)	Width (E)	Thickness (F)
		Inches	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
1	1	4 1/2	6	11/16	25/32	3/16	1/8	33/64	1/32	5/16	5/16
2	2	5 3/8	6 3/4	13/16	27/32	1/4	13/64	1/4	1/16	7/16	1/4
3	3	6 5/8	7 1/2	31/32	1 1/16	5/16	1/4	5/16	9/32	1/2	5/16
4	4, 5, 6	8	8 3/4	1 3/32	1 3/16	3/8	29/64	11/32	5/16	21/32	3/8

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE II.—*Sampling for visual and dimensional inspection A.Q.L. (approx.) = 1.5 percent defective*

Number of drill drifts in inspection lot	Number of drill drifts in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

Note—The Government inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL-STD-105

4.3 Sampling for lot acceptance tests.—A random sample of drill drifts shall be taken from each inspection lot in accordance with table III and shall be subjected to the tests specified in 4.6.

TABLE III.—*Sampling for lot acceptance tests*

Number of drill drifts in inspection lot	Number of drill drifts in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 to 800	25	1	2
801 and over	35	2	3

4.4 Inspection.

4.4.1 *Visual and dimensional inspection.* — Each of the sample drill drifts taken in accordance with 4.2 shall be visually and dimensionally inspected to verify compliance with this specification. Any drill drifts in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective drill drifts in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming drill drifts. The Government inspector shall again examine samples from such resubmitted lots to verify compliance with this specification.

4.6 Tests.

4.6.1 *Hardness.* — Each of the sample drill drifts taken in accordance with 4.3 shall be tested on a Rockwell testing machine to determine whether the drill drifts conform to the required Rockwell hardness specified in 3.4. The hardness readings shall be taken at no less than six different points within the area of the tapered section and handle end using a Rockwell testing machine. All readings shall be made after removing the outer surface to a depth of $\frac{1}{64}$ of an inch. In order that the Rockwell readings may be accurately obtained, care shall be exercised in supporting the drift so that no movement will occur and the surface under test will be horizontal.

5. PREPARATION FOR DELIVERY

5.1 Preservation. — Cleaning and preservation shall be in accordance with the manufacturer's standard practice

5.2 Packaging. — Commercial packages will be acceptable.

5.3 Packing. — The equipment shall be packed in substantial commercial containers of the types, size, and kind commonly used for the purpose in such manner as to permit acceptance and safe delivery to the designated point by common or other carrier at the lowest applicable rate.

5.4 Marking.

5.4.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name or trade-mark of the contractor, of such known character that the source of supply may be readily determined.

5.4.2 *Exterior containers.*—Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Style of drift required (see 1.2).

(c) Size of tool required (see applicable table).

(d) Whether preparation for immediate use, domestic storage, or overseas shipment is required (see 7.2.2).

6.2 Illustrations. — The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of drill drifts otherwise conforming to this specification.

6.3 Transportation description. — The applicable transportation description is as follows:

Pins, drift.

Carload minimum weight 40,000 pounds.
Truckload minimum weight 36,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification of the issue in effect on date of invitation for bids, forms a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specification:*

Military Specification:

MIL-P-15424—Packaging of Hand Tools; for Domestic or Overseas Shipment and Storage.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Preparation for delivery for domestic or overseas shipment.* — Drill drifts shall be cleaned, preserved, unit protected, packaged, packed, and marked in accordance with Military Specification MIL-P-15424.

MILITARY INTERESTS:

Army—O E Q Sig T
Navy—Sh A Or Y
Air Force.

U. S. GOVERNMENT PRINTING OFFICE: 1956

Copies of this specification may be purchased for 10 cents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.